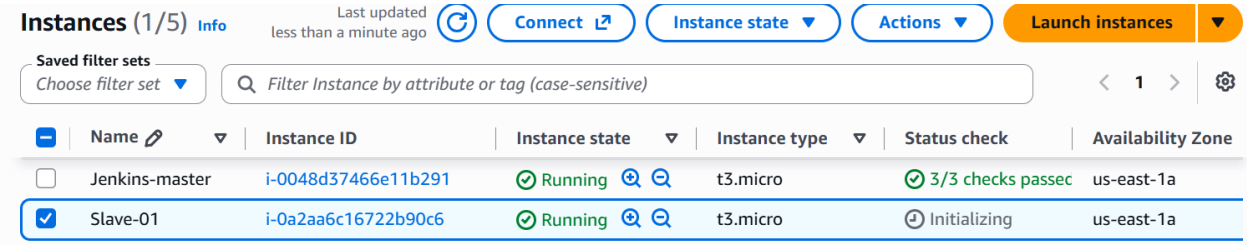
Configure 2 slave machines in the Jenkins master.




```
  libXtst-1.2.5-1.amzn2023.0.1.x86_64                   libbrotli-1.1.0-1.amzn2023.0.1.x8
  libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64             libpng-2:1.6.37-10.amzn2023.0.13.
  libxcb-1.17.0-1.amzn2023.0.1.x86_64                   pixman-0.43.4-1.amzn2023.0.4.x86_
  xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[root@ip-10-0-7-165 ~]#
```

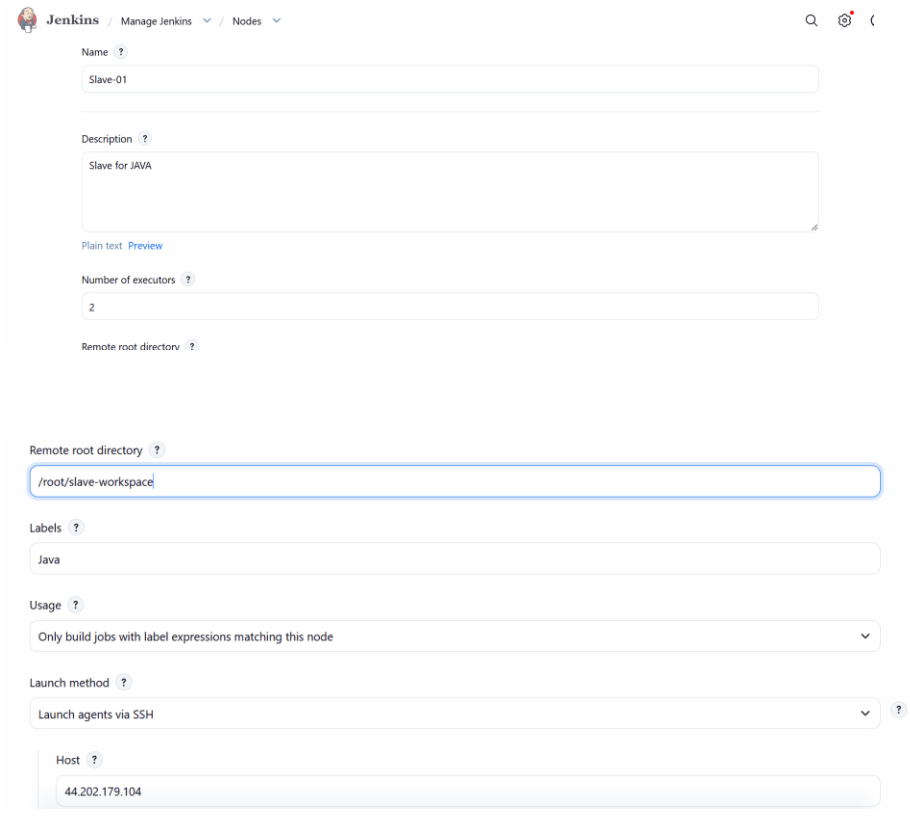
i-0a2aa6c16722b90c6 (Slave-01)

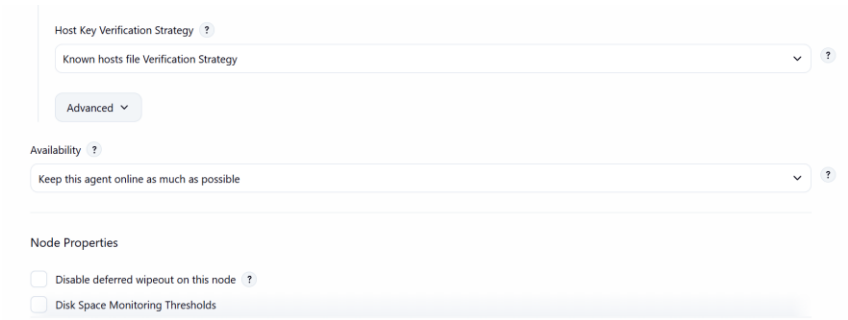
PublicIPs: 44.202.179.104 PrivateIPs: 10.0.7.165

Install java in slave-01

```
[root@ip-10-0-7-165 ~]# cd .ssh
[root@ip-10-0-7-165 .ssh]# ls
authorized_keys  id_rsa  id_rsa.pub
[root@ip-10-0-7-165 .ssh]# cat id_rsa.pub > authorized keys
cat: keys: No such file or directory
[root@ip-10-0-7-165 .ssh]# cat id_rsa.pub > authorized_keys
[root@ip-10-0-7-165 .ssh]# ll
total 16
-rw-r--r--. 1 root root  585 Sep 30 18:34 authorized
-rw-------. 1 root root  585 Sep 30 18:34 authorized_keys
-rw-------. 1 root root 2622 Sep 30 18:33 id_rsa
-rw-r--r--. 1 root root  585 Sep 30 18:33 id_rsa.pub
[root@ip-10-0-7-165 .ssh]# chmod 700 authorized_keys
[root@ip-10-0-7-165 .ssh]# ll
total 16
-rw-r--r--. 1 root root  585 Sep 30 18:34 authorized
-rwx------. 1 root root  585 Sep 30 18:34 authorized_keys
-rw-------. 1 root root 2622 Sep 30 18:33 id_rsa
-rw-r--r--. 1 root root  585 Sep 30 18:33 id_rsa.pub
```

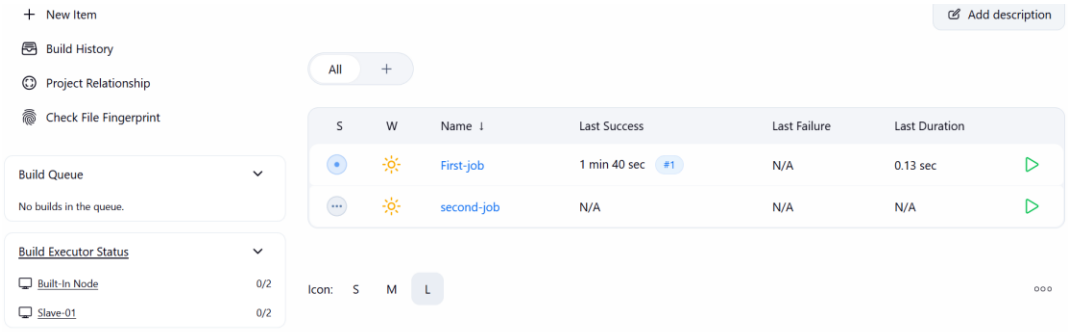
```
[root@ip-10-0-2-103 ~]# mkdir -p /var/lib/jenkins
[root@ip-10-0-2-103 ~]# rm -rf jenkins
[root@ip-10-0-2-103 ~]# cd /var/lib/jenkins
[root@ip-10-0-2-103 jenkins]# mkdir .ssh
[root@ip-10-0-2-103 jenkins]# cd .ssh/
[root@ip-10-0-2-103 .ssh]# ll
total 0
[root@ip-10-0-2-103 .ssh]# ssh-keyscan -H 44.202.179.104 >> /var/lib/jenkins/.ssh/known_hosts
[root@ip-10-0-2-103 .ssh]# ll
total 4
-rw-r--r--. 1 root root 564 Sep 30 18:40 known_hosts
[root@ip-10-0-2-103 .ssh]# pwd
/var/lib/jenkins/.ssh
[root@ip-10-0-2-103 .ssh]# ll
total 4
-rw-r--r--. 1 root root 564 Sep 30 18:40 known_hosts
[root@ip-10-0-2-103 .ssh]# chown jenkins:jenkins known_hosts
[root@ip-10-0-2-103 .ssh]# ll
total 4
-rw-r--r--. 1 jenkins jenkins 564 Sep 30 18:40 known_hosts
[root@ip-10-0-2-103 .ssh]# chmod 700 known_hosts
[root@ip-10-0-2-103 .ssh]# ll
total 4
-rwx------. 1 jenkins jenkins 564 Sep 30 18:40 known_hosts
[root@ip-10-0-2-103 .ssh]#
```



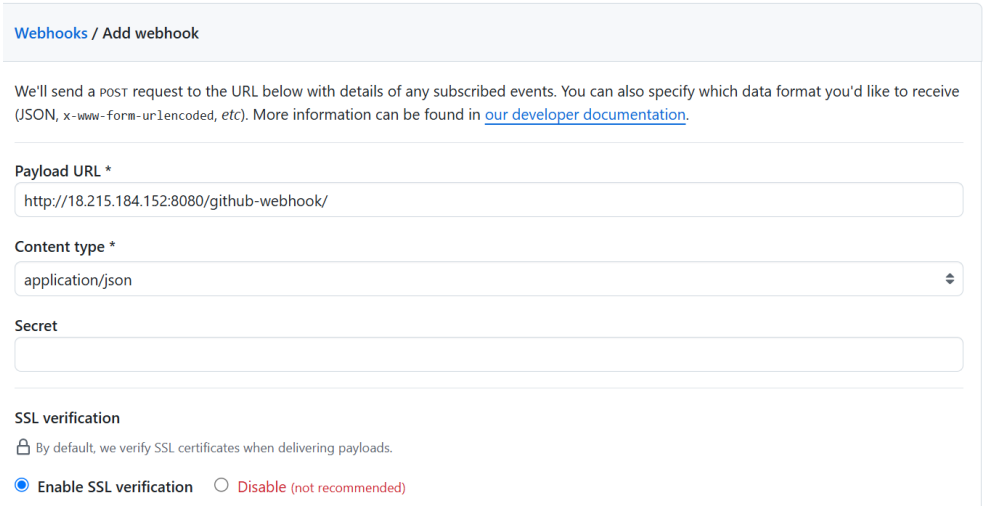



```
[root@ip-10-0-7-165 ~]# mkdir slave-workspace
[root@ip-10-0-7-165 ~]# cd slave-workspace/
[root@ip-10-0-7-165 slave-workspace]# pwd
/root/slave-workspace
[root@ip-10-0-7-165 slave-workspace]# ll
total 0
[root@ip-10-0-7-165 slave-workspace]# cd ..
[root@ip-10-0-7-165 ~]# ll
total 0
drwxr-xr-x. 2 root root 6 Sep 30 18:56 slave-workspace
[root@ip-10-0-7-165 ~]# chown ec2-user:ec2-user slave-workspace/
[root@ip-10-0-7-165 ~]# chmod 777 slave-workspace/
[root@ip-10-0-7-165 ~]# ll
total 0
drwxrwxrwx. 2 ec2-user ec2-user 6 Sep 30 18:56 slave-workspace
[root@ip-10-0-7-165 ~]# exit
logout
[ec2-user@ip-10-0-7-165 ~]$ ll
total 0
[ec2-user@ip-10-0-7-165 ~]$ pwd
/home/ec2-user
[ec2-user@ip-10-0-7-165 ~]$
```

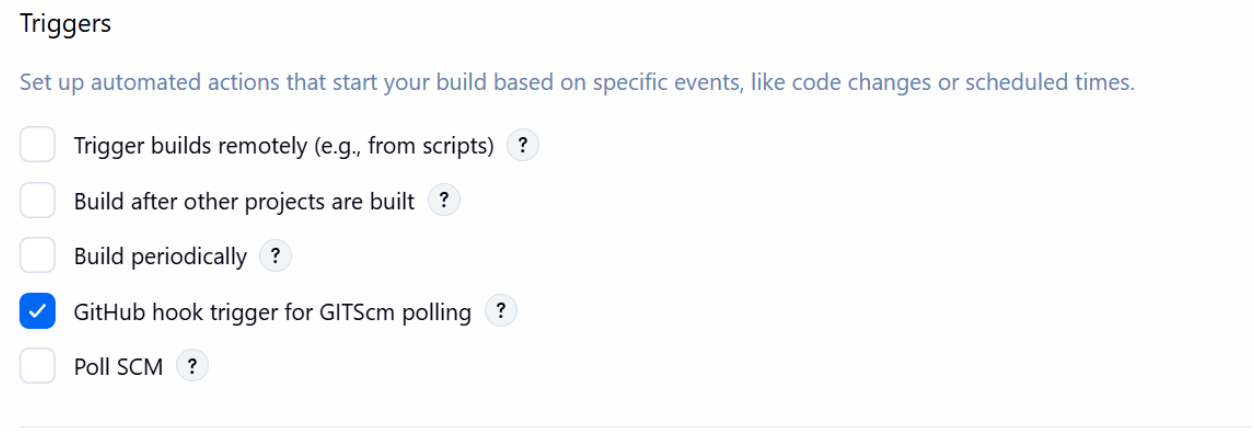
```
Remoting version: 3386.v353e57a_1b_ea_0
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```

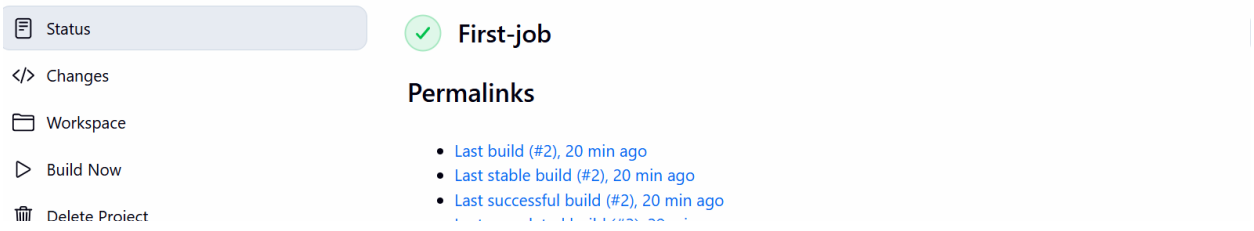


Configure webhooks to the Jenkins job.

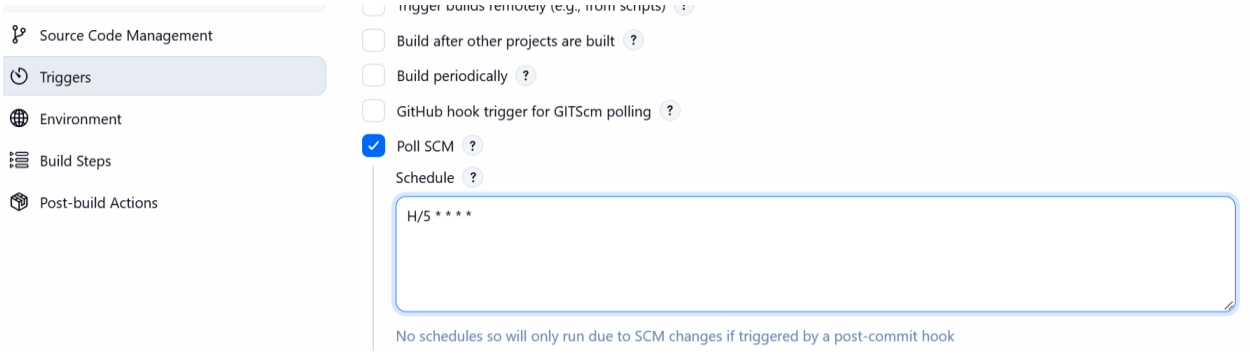


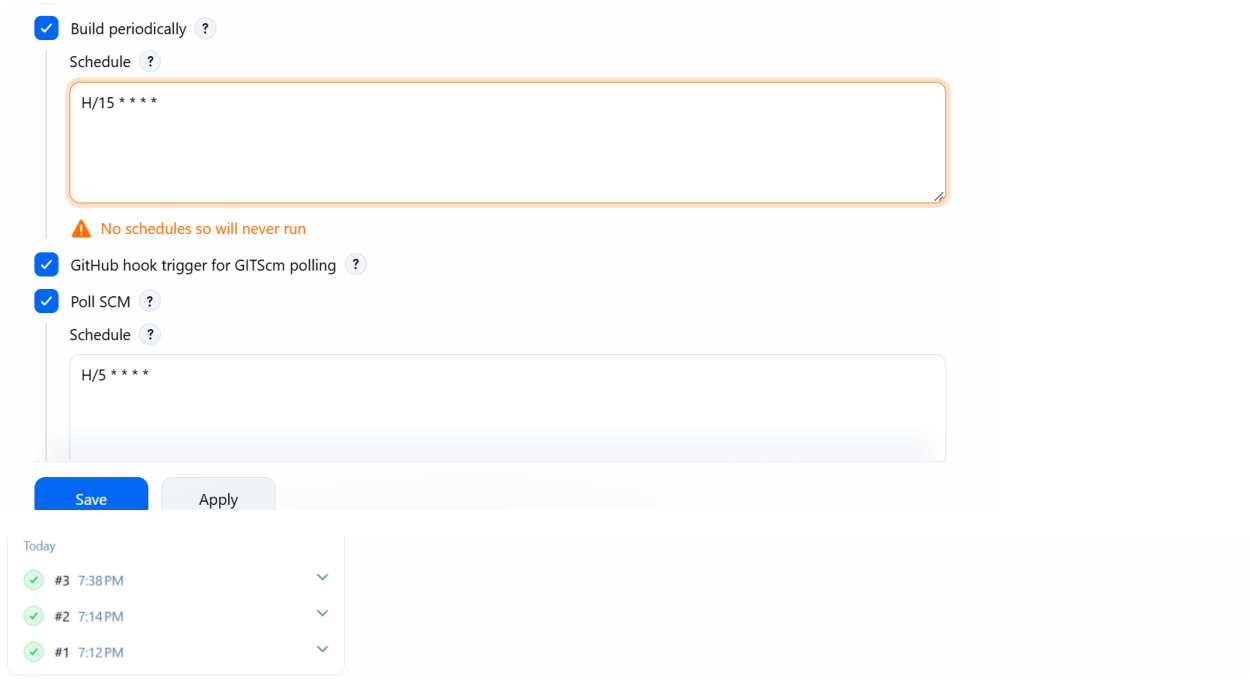
Add webhooks





Configure Poll SCM and Build Periodically options in the Jenkins job.





Take a backup of the Jenkins server using a bash-script.

```
[root@ip-10-0-2-103 .ssh]# cd /home/ec2-user
[root@ip-10-0-2-103 ec2-user]# vi jenkins_backup.sh
[root@ip-10-0-2-103 ec2-user]# cat jenkins_backup.sh
#!/bin/bash

BACKUP_DIR="/home/ec2-user/jenkins-backups"
DATE=$(date +%Y%m%d_%H%M%S)

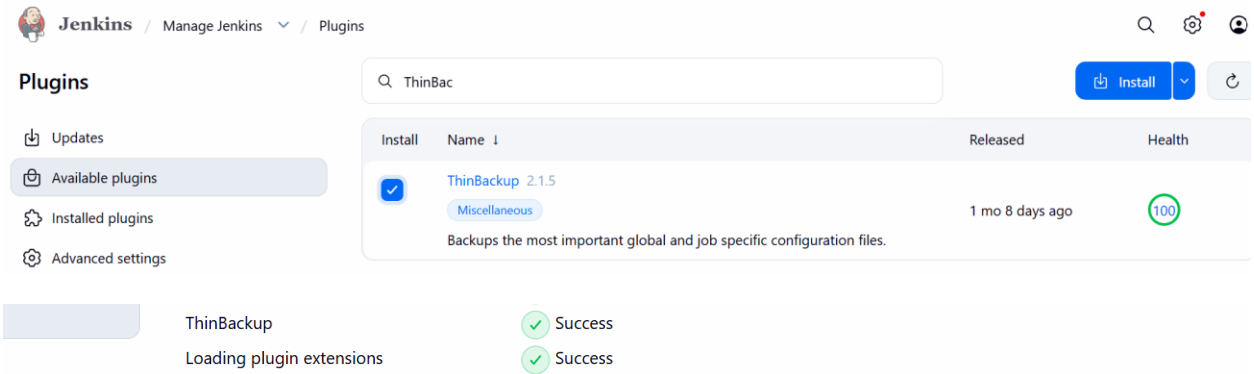
mkdir -p "$BACKUP_DIR"

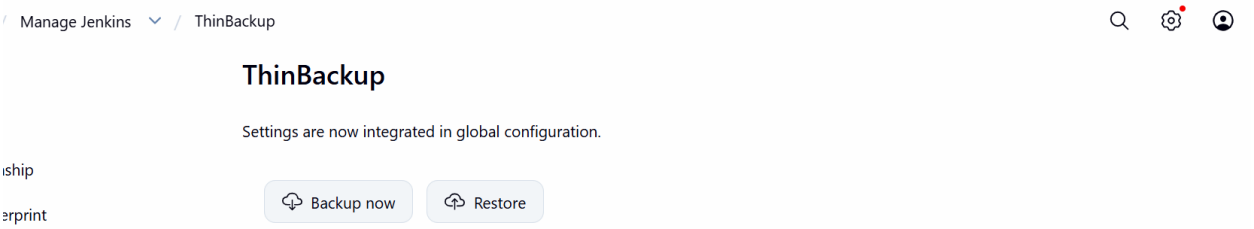
tar -czf "$BACKUP_DIR/jenkins_backup_$DATE.tar.gz" /var/lib/jenkins

echo "Jenkins backup completed successfully."
echo "Backup file: $BACKUP_DIR/jenkins_backup_$DATE.tar.gz"
[root@ip-10-0-2-103 ec2-user]#
```

```
[root@ip-10-0-2-103 ec2-user]# sudo chmod +x jenkins_backup.sh
[root@ip-10-0-2-103 ec2-user]# sudo ./jenkins_backup.sh
tar: Removing leading `/' from member names
Jenkins backup completed successfully.
Backup file: /home/ec2-user/jenkins-backups/jenkins_backup_20260930_194224.tar.gz
[root@ip-10-0-2-103 ec2-user]#
```

Take a backup of Jenkins using the ThinBackup plugin.






```
[root@ip-10-0-2-103 ec2-user]# sudo ls -lah /var/lib/jenkins/thinbackup
total 48K
drwxr-xr-x.  4 jenkins jenkins  64 Sep 30 19:50 .
drwxr-xr-x. 17 jenkins jenkins 16K Sep 30 19:49 ..
drwxr-xr-x.  5 jenkins jenkins 16K Sep 30 19:49 FULL-2026-09-30_19-49
drwxr-xr-x.  5 jenkins jenkins 16K Sep 30 19:50 FULL-2026-09-30_19-50
[root@ip-10-0-2-103 ec2-user]#
```

Set up a new Jenkins server and restore the backup taken in task 4.

```
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-09-30 20:03:22 UTC; 7s ago
   Main PID: 26611 (java)
      Tasks: 42 (limit: 1059)
     Memory: 362.1M
        CPU: 19.479s
     CGroup: /system.slice/jenkins.service
             └─26611 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=80>

Sep 30 20:03:17 ip-10-0-3-56.ec2.internal jenkins[26611]: [LF]>
Sep 30 20:03:17 ip-10-0-3-56.ec2.internal jenkins[26611]: [LF]> ************************************************************
Sep 30 20:03:17 ip-10-0-3-56.ec2.internal jenkins[26611]: [LF]> ************************************************************
Sep 30 20:03:17 ip-10-0-3-56.ec2.internal jenkins[26611]: [LF]> ************************************************************
Sep 30 20:03:22 ip-10-0-3-56.ec2.internal jenkins[26611]: 2026-09-30 20:03:22.399+0000 [id=31]        INFO        jenkins.InitReactorRunner>
Sep 30 20:03:22 ip-10-0-3-56.ec2.internal jenkins[26611]: 2026-09-30 20:03:22.439+0000 [id=24]        INFO        hudson.lifecycle.Lifecycl>
Sep 30 20:03:22 ip-10-0-3-56.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Sep 30 20:03:22 ip-10-0-3-56.ec2.internal jenkins[26611]: 2026-09-30 20:03:22.624+0000 [id=50]        INFO        h.m.DownloadService$Downl>
Sep 30 20:03:22 ip-10-0-3-56.ec2.internal jenkins[26611]: 2026-09-30 20:03:22.625+0000 [id=50]        INFO        hudson.util.Retrier#start>
Sep 30 20:03:27 ip-10-0-3-56.ec2.internal jenkins[26611]: 2026-09-30 20:03:27.505+0000 [id=71]        WARNING        h.n.DiskSpaceMonitorDe>

[ec2-user@ip-10-0-3-56 ~]$ aws --version
aws-cli/2.36.47 Python/3.9.25 Linux/6.18.51-120.162.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-10-0-3-56 ~]$
```

```
[root@ip-10-0-2-103 ~]# ls -lh /home/ec2-user/jenkins-backups/
total 145M
-rw-r--r--. 1 root root 145M Sep 30 19:42 jenkins_backup_20260930_194224.tar.gz
[root@ip-10-0-2-103 ~]#
```

```
india@Jaggu MINGW64 ~/Downloads
$ scp -i "Amazon-Linux-key.pem" ec2-user@ec2-18-215-184-152.compute-1.amazonaws.com:/home/ec2-user/jenkins-backups/jenkins_backup_20260930_194224.tar.gz .
Warning: Identity file Amazon-Linux-key.pem not accessible: No such file or directory.
ec2-user@ec2-18-215-184-152.compute-1.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
scp: Connection closed

india@Jaggu MINGW64 ~/Downloads
$ ls l *.pem
ls: cannot access 'l': No such file or directory
Amazon-Linux_key.pem  Linux-testing.pem  jenkins_key.pem

india@Jaggu MINGW64 ~/Downloads
$ scp -i "Amazon-Linux_key.pem" ec2-user@ec2-18-215-184-152.compute-1.amazonaws.com:/home/ec2-user/jenkins-backups/jenkins_backup_20260930_194224.tar.gz .
jenkins_backup_20260930_194224.tar.gz                                                                    100%  145MB 465.8KB/s   05:17

india@Jaggu MINGW64 ~/Downloads
$ ls -lh jenkins_backup_20260930_194224.tar.gz
-rw-r--r-- 1 india 197121 145M Oct  1 01:45 jenkins_backup_20260930_194224.tar.gz

india@Jaggu MINGW64 ~/Downloads
$ scp -i "Amazon-Linux_key.pem" jenkins_backup_20260930_194224.tar.gz ec2-user@100.59.7.171:/home/ec2-user/
The authenticity of host '100.59.7.171 (100.59.7.171)' can't be established.
ED25519 key fingerprint is SHA256:LeSnSJl9neOu5FCwKmKmUrb9dryIgTwmOH3+LU4sWgc.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '100.59.7.171' (ED25519) to the list of known hosts.
jenkins_backup_20260930_194224.tar.gz                                                                    100%  145MB 960.3KB/s   02:34
```

```
india@Jaggu MINGW64 ~/Downloads
$ ssh -i "Amazon-Linux_key.pem" ec2-user@100.59.7.171
        #_
    ~\_  ####_        Amazon Linux 2023
   ~~  \_#####\
   ~~     \###|
   ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
    ~~       V~' '->
     ~~~         /
       ~~._.   _/
          _/ _/
        _/m/'
Last login: Wed Sep 30 20:00:56 2026 from 18.206.107.27
[ec2-user@ip-10-0-3-56 ~]$ ls -lh /home/ec2-user/
total 145M
-rw-r--r--. 1 ec2-user ec2-user 145M Sep 30 20:21 jenkins_backup_20260930_194224.tar.gz
[ec2-user@ip-10-0-3-56 ~]$ sudo systemctl stop jenkins
[ec2-user@ip-10-0-3-56 ~]$ sudo rm -rf /var/lib/jenkins/*
[ec2-user@ip-10-0-3-56 ~]$ sudo tar -xzf /home/ec2-user/jenkins_backup_20260930_194224.tar.gz -C /
[ec2-user@ip-10-0-3-56 ~]$ sudo chown -R jenkins:jenkins /var/lib/jenkins
[ec2-user@ip-10-0-3-56 ~]$ sudo systemctl start jenkins
[ec2-user@ip-10-0-3-56 ~]$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-09-30 20:25:59 UTC; 5s ago
   Main PID: 27787 (java)
      Tasks: 49 (limit: 1059)
```

we implemented a Jenkins backup and restore process.

First, we created a Jenkins backup using a shell script that compressed the /var/lib/jenkins directory into a .tar.gz backup file.

We then copied the backup file from the original Jenkins server to the new Jenkins restore server using scp. On the restore server, we stopped Jenkins, removed the existing Jenkins data, extracted the backup into /, changed the ownership back to jenkins:jenkins, and started Jenkins again.

Finally, we verified that the Jenkins service was active and running, confirming that the backup was successfully restored.